

ZeroDrift — 5-Minute Selection Script (6 speakers)

Setup before you are called: Shivanand opens the app, loads `leo_pass_nominal`, real-time ON, ground-truth OFF, ready but PAUSED. Phone with the Netlify link open as backup. Everyone stands in speaking order, left to right. Speak slowly — slow feels professional.

0:00 – 0:40 · ARYAN (lead) — the hook

Good morning judges. We are Team ZeroDrift, and we solved ISRO problem 26169. Imagine two torches, kilometres apart, that must point exactly into each other's eyes — while both are shaking and one is moving at 27,000 km/h. That is laser communication. The data speed is amazing, but if the pointing fails, everything fails. ISRO asked for this pointing brain to be built and proven in software. We built it. Completely. And you will watch it work, live, right now.

[DO: Aryan nods to Shivanand → Shivanand presses START]

0:40 – 1:25 · VIVEK — what is on the screen

What you see is not an animation. It is a physical simulation of the real sky — stars, clouds, atmospheric shaking, and one blinking light called the beacon, riding a real ISS orbit computed from real orbital data. Our tracker sees only these noisy camera pixels — nothing else. Watch the left panel... it is searching... found it... and LOCKED. That took about one second, from completely cold.

1:25 – 2:10 · AASHNA — how it thinks

How does it know that dot is the beacon, and not a star? Simple idea: identity is not brightness — identity is the blink. The beacon blinks four times per second, like a secret handshake. A star can be brighter, but it can never blink right, so it scores zero and is rejected. Then our own small neural network — 497 parameters, trained by us, not downloaded — double-checks every lock. In 64 random test runs, it locked a wrong light exactly zero times.

2:10 – 2:50 · SHIVANAND — the live torture test (speaks while driving)

Let me make its life difficult. [DO: drag turbulence to maximum] That is heavy atmospheric shaking — see the beacon dancing. The tracker bends... but does not break. [DO: wait 3 seconds, point at the error graph] And here is the honest part. This whole time it never saw the answer. [DO: switch ground truth ON] The green marker is where the beacon truly was. Our tracker was right — and we only reveal the answer key after tracking, never during.

2:50 – 3:30 · PRANSHU — the engineering

This is finished engineering, not a concept. Five thousand lines of open-source Python. It processes nine megapixels every second and uses only twenty of every thirty-three milliseconds — on a normal laptop, no GPU, no cloud. Everything is public: our GitHub repository, a one-click desktop app, and a live web console — zerodrift-fsoc-pat.netlify.app — where you can replay this exact run on your own phone, right now.

3:30 – 4:05 · PALAK — the proof

And we measured everything the problem statement asks. Acquisition in 1.27 seconds. Lock held 98.3 percent of the pass. The beacon stayed inside the camera's view 100 percent of the time. 73 automated tests, all passing. And every number regenerates from one command — any judge can re-run our results and get the same answer. We do not ask for trust. We show proof.

4:05 – 5:00 · ARYAN — the close

Why does this matter? Without coarse alignment, this optical link works only 14.7 percent of the time. With ZeroDrift — 99.3 percent. That is the difference between a demo and a usable link. Today it aligns a simulated ISRO terminal; the same brain works for UAVs, ships and ground stations. Judges — the problem asked for algorithms proven in software without costly optics. We are the team that did not bring a promise. We brought the working system. Thank you — we welcome your questions.

Q&A — who answers what (one step forward, answer, step back)

Topic	Anchor	One-liner to build on
Beacon, blink, physics, orbits	Vivek	"The beacon is the terminal's identity light; 4 Hz because the camera's 30 fps can see it cleanly."
AI, wrong-target, detection	Aashna	"Gate is classical and strict; AI adds confidence — benchmarked 0.957 vs 0.900 classical."
Delay, control, mount	Shivanand	"Commands act 40 ms late, so a Smith predictor aims where the target WILL be."
Code, tests, app, website	Pranshu	"All open source; the site replays a logged run of this same engine."
Numbers, reproducibility	Palak	"Every figure comes from a logged run in the repo — I can show the file."
Why software-only, roadmap, anything unclear	Aryan	"The PS asks for validation without optical hardware — that is the point of a virtual test-range."

μrad question (someone will ask): ANYONE: "One micro-radian tilts the beam about 1 millimetre per kilometre. Our median error, 198 μrad, keeps the beacon deep inside the field for the fine-steering stage."

Golden rules

1. Never say "should work / basically / obviously." Say "**measured / watch / here is the proof.**"
2. If the app misbehaves: Shivanand says "let me show the recorded run of this same engine" → opens the Netlify link. No apologies, keep talking.
3. If asked something nobody knows: Aryan: "Honest answer — we haven't measured that yet; here is what we do know..." Honesty scores. Bluffing eliminates.
4. Rehearse 3 times with a phone timer. Target 4:45 — questions eat the rest.
5. Smile at LOCKED. It's your product's best moment.